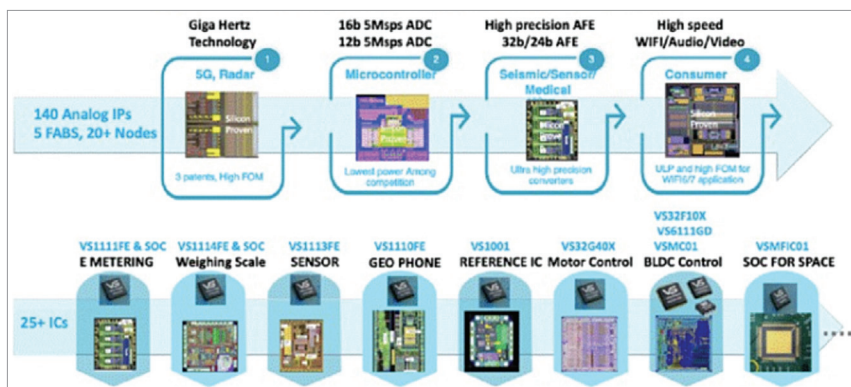




VerveseMI IPs and ICs portfolio



Pratap Narayan Singh, Founder Director, VerveseMI

time, observing how the hardware behaved during actual use.

In practice, no analogue system is perfect due to factors such as manufacturing variations, environmental conditions, ageing, power-supply fluctuations, reference-voltage drift, and signal-path noise. Engineers typically work to reduce these issues, but they cannot eliminate them entirely.

“But by integrating a machine learning-based system, we focus on continuously observing these deviations as they occur, learning from them dynamically, and building a digital model of the analogue behaviour, including its imperfections. We then use this model to correct errors in real time, adjusting system behaviour to improve accuracy, stability, and overall performance,” says Pratap.

The company has developed a broad portfolio of more than 140 semiconductor intellectual property (IP) blocks, which it has been

licensing internationally over the past eight years. These IPs cover a wide range of analogue and mixed-signal functions that form the foundation of modern signal chains, including clocking systems, ADCs, DACs, variable-gain amplifiers (VGAs), programmable-gain stages (PGs), and precision filters.

Speaking about the design challenges, the company has faced, Pratap says, “Designing analogue systems is not just about architecture or circuit design. The real challenge begins when these chips are deployed in the field, where customers need to determine whether a fault lies in the chip or elsewhere in the system.”

At production scale, testing millions of units quickly becomes a bottleneck, making traditional methods impractical. In critical applications such as automotive and aviation, even a single failure is unacceptable, necessitating robust diagnostics, redundancy, and high

reliability. Unlike digital systems, analogue designs operate at the transistor and signal level, making both testing and manufacturing significantly more complex.

The company operates design and R&D centres in Greater Noida and Bengaluru while using external semiconductor fabrication partners, consistent with the fabless model.

“We are already engaged with semiconductor fabs working on advanced technology nodes and are actively exploring future development, including components at the 4nm scale. If a customer moves from nodes such as 12nm to 8nm or 5nm, we align our development accordingly and build solutions to meet those specific needs,” adds Pratap.

In terms of academic and institutional collaboration, the company is engaged with select IITs, primarily for training and capability development.

—Nidhi Agarwal

3 AMIMOTION'S WORLD-SMALLEST UWB TAGS THAT GIVE FACTORIES A DIGITAL NERVOUS SYSTEM FOR PHYSICAL AI

A Bengaluru startup is turning factories into intelligent spaces with tiny UWB tags, AI-powered visibility, and real-time tracking that can automate workflows, improve safety, and prepare industries for Physical AI.

Factories and warehouses have become smarter with automation, connected machines, and AI systems. Yet a major gap remains: industries still lack

complete visibility into physical environments. Workers spend valuable time searching for trolleys, scanners, tools, and materials, while many workflows still rely on

repetitive barcode or RFID scanning. Safety also remains a challenge because worker locations are not always known during emergencies. While AI has transformed digital